

PRDT: Research as a Quant Problem

An AI-Native Research Intelligence Framework

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PRDT program (internal research intelligence + public P-Research layer)

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Abstract

We present PRDT, an AI-native research intelligence framework built on the thesis that the research frontier can be treated as a market: papers are data points, method families are factors, surveys are portfolios, and audits are risk control. PRDT runs a deterministic five-agent loop — open source, frontier paper, professional report, real-time information, and analysis/synthesis — over domain adapters (a six-pillar frontier-AI taxonomy and a fixed-income domain), producing a research graph with association, contradiction, and hypothesis engines. Its public layer, P-Research, is a continuously-running product that sweeps arXiv weekly into an append-only corpus (503 papers, 2023-2026), verifies full texts (45 records at 99% tag-match rate), tracks citations and group-level activity, and regenerates an audited, PDF-compiled survey from the corpus. We report the engineering that makes the loop trustworthy: truncated response salvage, dual-source full-text verification, keyless citation fallback, and an automated citation audit that has caught 24 mechanical errors (21 truncated record ids, 3 wrong external arXiv ids) before any human review. We argue that deterministic, evidence-chained research infrastructure is a missing layer between paper databases and LLM research agents, and discuss the open problems: full-text coverage, citation lag, the closed frontier, and the human gates that remain.

Paper 2 — §1 Introduction (draft v0.1)

1 1.1 The research-as-market thesis

Quantitative finance treats markets as data and converts raw price streams into factors, portfolios, backtests, and risk reports. This paper proposes that the research frontier admits the same treatment, with a materially better starting position: the market data — papers, citations, code, and institutional affiliations — is public by default.

Under this lens, the mapping is direct. A paper is a data point with high-dimensional features (methods, benchmarks, models, citations). A method family is a factor whose momentum can be measured week over week. A survey is a portfolio: claims are positions, each backed by evidence and sized by confidence. A citation audit is risk control. And the researcher's alpha is the same as a quant's: seeing structure earlier and more reproducibly than the market does.

The thesis is most tractable in CS/AI. Papers ship code, so "backtesting" a claim means running the released repository. The feedback loop is weekly — arXiv updates every day. And unlike trading, the evidence chain is free: every claim can be traced from survey text to database record to original manuscript, with no proprietary data in between.

2 1.2 Why deterministic infrastructure (and not another agent)

LLM research agents are increasingly capable, but they inherit the stochasticity of their substrate: two runs produce two answers, and "reasoning" is not auditable in the sense risk control requires. PRDT takes the opposite design position: ****the pipeline itself contains no LLM calls****. Ingestion, structuring, verification, clustering, and auditing are deterministic programs with schema-validated outputs. LLMs are — by design — kept outside the evidence chain, reserved for the human-readable synthesis layers where judgment, not reproducibility, is the goal.

This buys three properties that agents cannot yet claim:

1. **Auditability**: every claim re-derives from raw inputs.
2. **Reproducibility**: the same corpus state produces the same survey.
3. **Honest signal strength**: abstract-level tags are labeled as such; only full-text-verified records may back survey claims.

3 1.3 The two-track architecture

PRDT ships as two coordinated systems rather than one monolith:

- **PRDT (internal)**: the five-agent research loop, research graph, and

hypothesis/experiment engines — the private "strategy" layer, where hypotheses, contradictions, and internal conclusions live.

- **P-Research (public)**: the product layer — a weekly-sweeping corpus

pipeline with verification, citation tracking, group radars, a high-dimensional paper matrix, and the living survey it generates.

The echo between them mirrors the quant pattern of private funds and public indices: the public layer thickens the data, attracts collaborators, and earns reputation; the private layer consumes that corpus to do deeper, opinionated research. Neither competes with the other; they compound.

4 1.4 Contributions

1. A five-agent, domain-agnostic research intelligence framework with a research graph and domain-aware association/contradiction/hypothesis engines (Section 3).
2. Robustness engineering for the public scholarly data layer: truncated-response salvage, dual-source full-text verification, keyless citation fallback, fail-fast rate-limit circuits (Section 4).
3. An evaluation of the running system over a 503-paper corpus, including a 99% tag-match verification rate and an automated audit that has caught 24 mechanical citation errors pre-review (Section 5).
4. Evidence that the framework's flagship output — a living, audited survey — is itself a research contribution, not just a product (Section 6).

5 1.5 Scope and honesty

This is a systems paper about infrastructure, not a claim that research is solved. The system’s limitations are enumerated in Section 7: the corpus sees only the open literature; full-text coverage is a fraction of the database; recent papers suffer citation lag; and the final publishing gates — human review and endorsement — remain, by design, outside the machine.

Paper 2 — §2 Design Principles (draft v0.1)

PRDT is a systems position paper as much as a system description. Four principles separate it from both “paper database + LLM wrapper” tools and from pure agent frameworks.

6 2.1 Deterministic and auditable

Every transformation in the evidence chain — arXiv ingestion, structuring, verification, clustering, auditing — is a deterministic program. The structurer is a keyword/regex engine with word-boundary rules for short acronyms (e.g., “cot” must not match inside “scotland”); the verifier re-runs the same taggers over full text; the auditor re-derives every quoted number from the database at run time. No LLM call sits anywhere inside the chain, because an LLM in the chain would make two properties unattainable:

- **Re-derivability:** anyone can regenerate a claim from raw inputs and

get the same answer.

- **Blame assignment:** when a claim is wrong, the audit names the exact

mechanical failure (truncated id, wrong external id, stale number) — something a stochastic agent cannot do.

7 2.2 Append-only evidence, atomic verification

The corpus (`data/papers.jsonl`) is append-only: ingestion never rewrites history. Verification metadata merges into records via atomic file replacement (temp file + `os.replace`), so a concurrent sweep can never read a half-written database. The invariant: the database grows, but any past state is reconstructible from the sweep logs and the raw arXiv cache.

8 2.3 Living artifacts, regenerated claims

The survey is not a frozen PDF: it is assembled (`presearch assemble`), audited (`presearch audit`), and compiled (`presearch latex`) from the current corpus state on every weekly run. Quoted numbers are therefore *claims about the corpus at time t^* , and the audit fails loudly when the corpus moves out from under a claim. Static surveys rot; this one is designed to contradict itself loudly rather than silently.

9 2.4 Honest signal strength

Evidence is labeled by its verification state, and only the strongest state backs survey claims:

1. **abstract-level** — query-of-origin pillar + keyword tags (weak signal).
 2. **full-text-verified** — ar5iv/arXiv-HTML full text re-tagged, method overlap recorded (strong signal).
 3. **audited** — the claim-level check (record ids exist, numbers match).
- The README and every digest carry the disclaimer that abstract-level evidence is a research signal, not a fact. This is not modesty; it is risk control — the research-market equivalent of marking a position’s liquidity class.

Paper 2 — §3 Architecture (draft v0.1)

10 3.1 The five-agent research loop

PRDT’s internal layer is organized as five cooperating agents over a shared, schema-validated evidence format:

Agent	Input class	Output
Open-Source Project Agent	repositories	reproducibility/integration triage
Frontier Paper Agent	arXiv papers	thesis, method, replication plan, targets
Professional Report Agent	institutional reports	fact/opinion/forecast separation
Real-Time Information Agent	events	impact ranking, affected conclusions
Analyze & Synthesis Agent	fused evidence	associations, contradictions, hypotheses, experiments

Each agent’s output is validated against a required-field schema before it may enter the pipeline (`REQUIRED_FIELDS`), and every agent is deterministic — the paper agent, for example, converts an arXiv entry into evidence with pure keyword/regex rules rather than an LLM call.

11 3.2 Research graph and domain-aware engines

The synthesis agent builds a research graph whose nodes are evidence, associations, contradictions, and hypotheses, and whose edges are typed (`SUPPORTS`, `CONTRADICTS`, `TESTS`, ...). Three engines produce the graph’s intellectual content:

- **Association engine:** clusters evidence by shared method tags; in the `ai` domain, cross-pillar convergence (a method family spanning ≥ 2 pillars with ≥ 3 papers) is the primary association rule.
 - **Contradiction engine:** weakens seeded conclusions when new evidence matches a per-conclusion challenge lexicon; the `pficc` domain instead consumes explicit `affected_existing_conclusions` metadata.
 - **Hypothesis engine:** converts associations into survey claims with verification experiments attached (full-text review, lineage checks).
- Domain-awareness is the design point: the same engines produce fixed-income factor hypotheses for `pficc` and frontier-research claims for `ai`, from domain adapter definitions rather than hardcoded text.

12 3.3 Domain adapters

A domain adapter is a declarative bundle: taxonomy, source priorities, evaluation metrics, integration targets, and (for **ai**) per-pillar arXiv query plans. Two adapters ship:

- **pficc** — China fixed-income multi-factor research (rates/credit/portfolio/trading taxonomy).

- **ai** — the six-pillar frontier taxonomy (Reasoning / Agents / Efficiency / RL-Alignment / Multimodal / Quant×AI).
Adding a domain is a configuration act, not a code change; the registry (`get_domain`) resolves adapters by name.

13 3.4 The internal↔external echo

PRDT is deliberately two tracks that compound:

- **PRDT (private)**: hypotheses, contradictions, experiment queues, internal conclusions — the strategy layer.

- **P-Research (public)**: the same evidence pipeline productized — weekly sweeps, verification, citations, group radars, a paper matrix, digests, and the living survey.
The echo is mechanical, not metaphorical: PRDT ingests the public corpus verbatim (`prdt ingest presearch`, 503 records) and appends a "P-Research corpus echo" section to its own weekly digest; P-Research's methodology (structurer, audit, connectors) originates in PRDT. This is the research-market version of the private-fund/public-index pattern: the public layer thickens the data and earns reputation, the private layer spends that reputation on opinionated research.

Paper 2 — §4 Retrieval & Robustness Engineering (draft v0.1)

The public scholarly data layer is friendly in principle and hostile in practice: rate limits, truncated responses, moving API fields, and coverage gaps. This section documents the engineering that keeps the loop running — each item below is a failure we actually hit, and the mechanism that survived it.

14 4.1 Truncated-response salvage (arXiv)

Flaky networks truncate large arXiv Atom responses mid-stream (`http.client.IncompleteRead`, e.g. 31KB of 41KB delivered). Retries alone are insufficient when truncation is bursty. The connector salvages the partial payload: the `IncompleteRead.partial` bytes are parsed for complete `<entry>...</entry>` blocks (each block is wrapped in a minimal namespace-bearing feed and parsed independently), so a truncated response yields most of its papers instead of nothing. On the first full sweep, 5 of 6 pillar queries were salvaged this way.

15 4.2 Dual-source full-text verification

Full-text verification prefers ar5iv (LaTeXML HTML) and falls back to arXiv’s native HTML rendering for papers ar5iv has not converted. Two pitfalls required explicit handling:

- **Abs-page masquerade:** for unconverted papers, both sources can

return the arXiv abstract page (HTTP 200, 40KB of navigation HTML).

Detection: real paper HTML carries `ltx_` markers; abs pages do not.

False positives are rejected, not ingested.

- **Truncated HTML:** partial payloads ≥ 50 KB are still usable for text

extraction (HTML parsing is lenient); smaller partials trigger retry.

The verification gate re-runs the deterministic taggers over full text

and records the method-tag overlap against the abstract-level record.

On the first 45 verified papers the match rate was 99% (81/82),

empirically validating the abstract-level structurer.

16 4.3 Keyless citation fallback (OpenAlex)

Semantic Scholar’s anonymous pool is frequently saturated (we observed sustained HTTP 429). The citation layer therefore treats OpenAlex — fully keyless, with citation counts and institutional metadata — as the default source, and S2 as an optional keyed upgrade. Two data-engineering lessons surfaced:

- **Field location:** `raw_affiliation_strings` is not a top-level work

field and is not in the `select` whitelist; it lives inside each

`authorships` entry. Department-level group tracking (e.g., "HKU

Computer Science" vs "HKU" at large) is only possible after extracting it from the right place.

- **Cache typing:** a cache loader that accepted only dicts silently

discarded every cached list — turning a cache hit into a network storm

and a 429 cascade. Accepting both container types fixed it.

Citation lag is handled at the scoring layer: the novelty-citation blend

falls back to method-overlap novelty alone when citation data is absent, so recent papers are never penalized for being too new to be cited.

17 4.4 Fail-fast circuits

Sustained 429s are not retried into the ground: the connector breaks

after two consecutive rate-limited attempts and a group-sweep circuit

breaker stops remaining watchlist entries, preserving the polite pool

for the next scheduled run. Merge semantics guarantee that a failed

refresh keeps the previous data — the system degrades to stale-but-true rather than empty.

18 4.5 The audit as the closing gate

presearch audit closes the loop with six mechanical checks: record-id existence, verification status of cited records, arXiv id format and membership, headline corpus numbers, the §3 trend table recomputed from the verified subset, and hard claim checks (three quoted numbers re-derived from the database). It has caught 24 mechanical errors in this project’s own draft — 21 truncated record ids and 3 wrong external arXiv ids — before a single human reviewed the paper. The lesson we draw: in a pipeline that distills thousands of records into prose, mechanical error is the base rate, and the only defense is a machine that checks the machine.

Paper 2 — §5 Evaluation (draft v0.1)

Evaluation of a research-intelligence framework is unusual: the ”test set” is the live scholarly frontier, and the ”labels” are the paper records themselves. We evaluate along four axes: corpus scale, signal quality, mechanical error detection, and the usefulness of downstream artifacts. All numbers below are re-derivable from the public repository.

19 5.1 Corpus scale and coverage

The P-Research corpus holds **503 papers** (2023-2026), built from weekly sweeps (21-day windows) plus non-overlapping historical backfills (0-360, 360-720, 720-1080, 1080-2160 days). Year coverage: 2023 (140), 2024 (112), 2025 (131), 2026 (120). Six pillars are balanced at the abstract level (77/75/70/64/60/59 at first gate; quant pillar later grew to 95+ via backfill).

20 5.2 Signal quality: the 99% bridge

Full-text verification spans **92 records across all six pillars** (45/10/10/9/9/9). On the first 45 verified lead-pillar records, the method-tag match rate between abstract-level and full-text tagging was **99% (81/82)** — empirical validation that the deterministic structurer’s abstract-level signal is a reliable proxy where full text has not yet been pulled. This number is the bridge that licenses §5.8 of the survey: the abstract-level convergence table stands in where verification has not reached.

21 5.3 Mechanical error detection

The automated citation audit runs six checks on every survey regeneration (record-id existence, cited-record verification status, arXiv id format and membership, headline corpus numbers, trend-table recomputation, and three hard claim checks). In production it has caught:

- **21 truncated record ids** in the draft (15-char prefixes of 20-char database ids), fixed by prefix resolution;

- **3 wrong external arXiv ids** that resolved to unrelated papers

(regular languages, AI accountability, quantum noise), removed.

Both failures predated any human review. The base-rate claim we make is specific: in a pipeline that distills hundreds of records into prose, *mechanical citation error is the base rate*, and a deterministic auditor is the cheapest defense against it.

22 5.4 Downstream artifacts

- **Research-group tracking:** 7 institutional groups across HK

universities (26/14/12/10/9/5/1 papers) plus the HKUDS repo radar — 92 repositories, 355K stars, 54 active in 2025+, classified onto the six-pillar taxonomy with a venue-year evolution curve (2023: 20 paper-repos, then a 2024-2025 shift toward agentic products).

- **High-dimensional matrix:** 503 papers in a TF-IDF space (4,000

terms), cosine retrieval, and k-means over 12 data-driven research directions with pillar mixes and year distributions.

- **Citation layer:** 55 records with citation counts via the keyless

OpenAlex source; novelty-citation blend scoring with citation-lag fallback.

23 5.5 Limitations of this evaluation

Numbers are corpus statements, not field-wide claims; recent papers are over-represented by weekly windows; pillar assignment follows the query of origin; and the closed frontier (commercial systems) is invisible to the open-literature corpus by construction.

Paper 2 — §6 The Survey as a System Output (draft v0.1)

A framework is evaluated by what it produces. PRDT’s flagship output is the living survey (“The Compute-Allocation Frontier”), and the survey is worth examining as an artifact of the system rather than as a standalone document.

24 6.1 Assembly from evidence

The survey is not written once; it is assembled. Seven section files are concatenated by `presearch assemble`, converted to LaTeX by a deterministic markdown-to-TeX converter, and compiled to a 13-page PDF with zero errors. Every quoted number in the draft is a claim about the corpus at generation time, and the audit re-derives each one.

25 6.2 The claim pipeline

The survey’s intellectual content flows through three gates:

1. **Evidence** — structured records with verification state.
2. **Claim** — a sentence in a section that cites **evg-*** records.
3. **Audit** — a machine re-check that the cited records exist, are verified where claims require it, and that numbers match.

This pipeline is what turns ”a survey” into ”a survey with an evidence chain” — the property that makes the document updateable without rotting.

26 6.3 What the survey demonstrates about the framework

- The compute-allocation lens (train vs. test time; model vs.

environment; representation vs. reward) is the survey’s organizing claim — and it is itself a hypothesis the framework generated from the corpus’s cross-pillar convergence signals.

- The survey’s §5.8 verified-subset matrix shows the framework’s

self-correction loop: an abstract-level claim (method families converge across pillars) was promoted to a full-text-verified table (n=92) within one iteration.

- The survey ships with its own honest-limitations section, including the

benchmark-visibility asymmetry (§6.2 of the survey): hard benchmarks (AIME, GPQA) hide in full text and are systematically under-counted by abstract-level mining — a finding that emerged from the verification pipeline itself.

27 6.4 Toward a general claim

We believe the pattern generalizes beyond this survey: any research program that maintains a corpus can produce *living documents* — surveys, landscape reports, group radars — assembled from the same evidence layer, audited by the same machine. The survey is the proof of concept; the framework is the claim.

Paper 2 — §7 Limitations & Future Work (draft v0.1)

28 7.1 Corpus limitations

- **Abstract-level assignment:** pillar labels follow the query of

origin; method tags come from keyword matching. The 99% match rate (§5.2) bounds but does not eliminate this noise.

- **Coverage:** full-text verification reaches 92 of 503 records. The

remaining evidence is labeled as signal, not fact.

- **Closed frontier:** commercial systems publish no full text; the corpus sees them only through third-party evaluation papers.

- **Citation lag:** recent papers lack mature citation graphs; the novelty-citation blend falls back to method-overlap novelty, which is a proxy, not a substitute.

29 7.2 System limitations

- **Determinism is a floor, not a ceiling:** the pipeline cannot formulate genuinely new hypotheses on its own — the hypothesis engine instantiates templates over association signals. LLM-assisted hypothesis *generation* (outside the evidence chain) is future work.
- **Human gates remain, by design:** reviewer judgment, endorsement, and publication decisions are outside the machine. This is a feature of the evidence chain, not a defect — but it caps autonomy.
- **Rate-limit fragility:** the polite pools of arXiv/OpenAlex/GitHub throttle batch runs; the system degrades to stale-but-true, which is acceptable for weekly cadence but not for burst demand.

30 7.3 Future work

1. **Plugin ecosystem:** pillar/connector/watchlist manifests (`presearch_pillars.json` ships; watchlist and connector manifests are next) so external researchers extend the taxonomy without forks.
2. **Direction momentum:** time-series the weekly cluster sizes and method migrations to measure research-direction momentum and overheating — the research-market analogue of factor momentum and crowding.
3. **Full-text coverage:** extend verification toward the full corpus; store extracted full-text claims for citation-level retrieval.
4. **LLM in the loop, outside the chain:** use LLMs to *propose* hypotheses and prose, then route their outputs through the same deterministic audit — the research-market version of a risk overlay.
5. **Venue evaluation:** run the living-survey pattern on a second domain (e.g., quant×AI as its own survey) to test the generality claim of §6.4.

31 7.4 Closing remark

We built PRDT on a specific wager: that in research, as in markets, the durable edge is infrastructure that sees structure earlier and can prove it. The survey is the first portfolio; the pipeline is the fund; the audit is the risk desk. Whether the wager pays is now an empirical question — and the machine will keep measuring it every Monday.